

HARDWARE AND COMPONENTS BASICS

Day 1: OVERVIEW OF HARDWARE COMPONENTS

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1. Labeled Diagram of a Desktop PC



2. HDD vs SSD Comparison

FEATURE	HDD	SSD
Speed	Slower because it uses a spinning disk.	Faster because it uses flash memory.
Durability	Less durable because it has moving parts.	More durable because it has no moving parts.
Shock Resistance	Can be damaged more easily by strong shocks.	Better resistance to shocks and movement.
Storage Capacity	Usually available in large capacities at lower cost.	Large capacities are available, but higher capacities can cost more.
Common Use	Useful for desktops, backups and storing large files.	Common in laptops, gaming PCs and as the main system drive.
Noise	Can make some noise while working.	Works silently.
Boot / Loading Time	Generally takes more time to start and load files.	Usually starts the computer and loads files much faster.

Conclusion: SSD is faster, quieter and more resistant to shocks than HDD. HDD is still useful when a large amount of storage is needed at a lower cost.

3. ROLE OF CPU, RAM AND GPU IN GAMING

CPU: The CPU is the main processor of a computer. In a game like PUBG or Free Fire, it processes game instructions, player actions and other calculations. A good CPU helps the game respond properly and run smoothly.

RAM: RAM is the temporary memory of the computer. When a game is running, it keeps game data and other required information ready for quick access. Having enough RAM helps reduce slowdowns, especially when other applications are open at the same time.

GPU: The GPU is responsible for creating and displaying the game's graphics. It processes visuals such as characters, maps, textures, lighting, and effects. A good GPU helps the game run with better graphics and smoother frame rates.

4. Hardware Upgrades for FIFA

If I want to upgrade my PC to run the latest version of FIFA smoothly, I would first check the game's system requirements. After that, I would upgrade the components that are below the recommended requirements.

1. GPU: I would consider upgrading the GPU because it is mainly responsible for game graphics. A suitable GPU can improve graphics quality and help maintain smoother FPS during gameplay.

2. RAM: I would upgrade the RAM if the current amount is not enough. More RAM gives the game enough memory to work with and can help when other programs are running in the background.

3. CPU: I would upgrade the CPU if the existing processor is old or does not meet the game's requirements. A better CPU can handle game calculations and player actions more efficiently.

4. SSD: I would use an SSD for the operating system and game because it provides faster loading and startup times. It can make opening the game and loading different parts of it quicker.

5. Power Supply (PSU): If I install a more powerful GPU or CPU, I would also check the power supply. The PSU should be powerful enough to safely provide electricity to all the components.

Final Point: For a gaming PC, the CPU, RAM, GPU and storage all have an important role. I would not upgrade every component without checking first. I would compare my current hardware with the game's recommended requirements and then upgrade the components that need improvement.